The Very Long-Run Costs of Financial Crises

Discussion of Müller, Xu and Lehbib

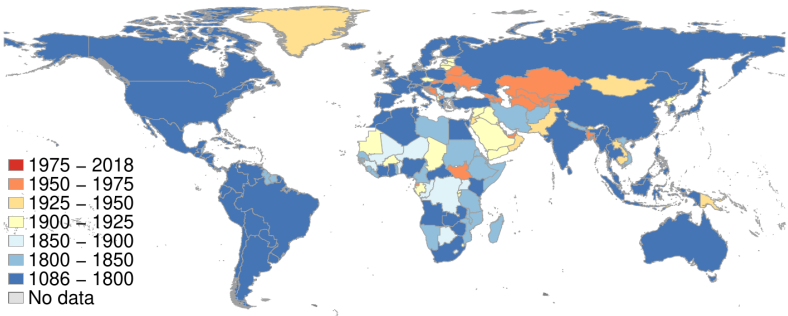
Sanjay R. Singh
FRB San Francisco and UC Davis

NBER International Finance and Macroeconomics, Fall 2026

Views are mine and not necessarily those of the Federal Reserve Bank of San Francisco or the Federal Reserve System.

A remarkable public good

Figure A.1: First Year in Dataset, By Country



239 economies
46 macro indicators

Congrats and thank you on assembling this detailed collection

Outline

1. Why do we care about long horizon IRFs?
2. How far the evidence can take us?

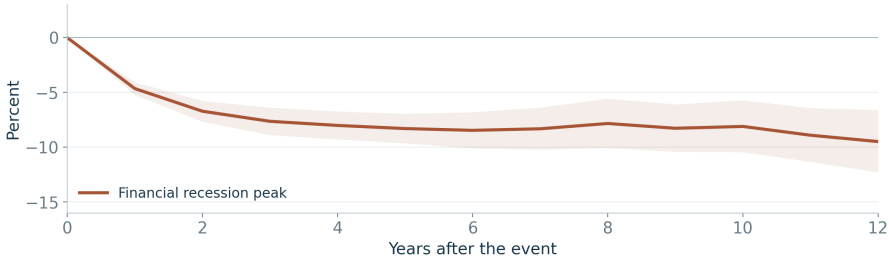
Financial recessions leave persistent losses

$$y_{i,t+h} - y_{it-1} = \alpha_{i,h} + \theta_h F_{it} + \sum_{\ell=1}^{p} \gamma_{h,\ell} \Delta y_{i,t-\ell} + u_{i,t+h}.$$

Financial recessions leave persistent losses

$$y_{i,t+h} - y_{it-1} = \alpha_{i,h} + \theta_h F_{it} + \sum_{\ell=1}^{p} \gamma_{h,\ell} \Delta y_{i,t-\ell} + u_{i,t+h}.$$

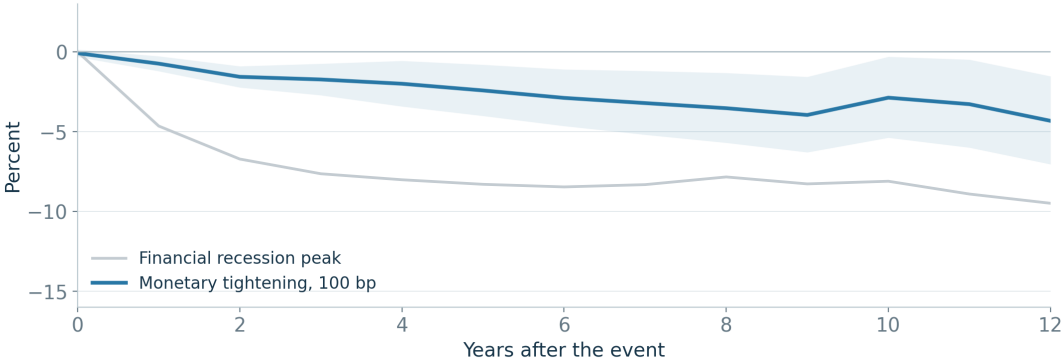
"Financial recessions": Jordà Schularick & Taylor (2013) *When Credit Bites Back* (2013) classification.



GDP is about 9.5% lower at year twelve.

Four annual growth lags; country fixed effects. World war windows excluded.

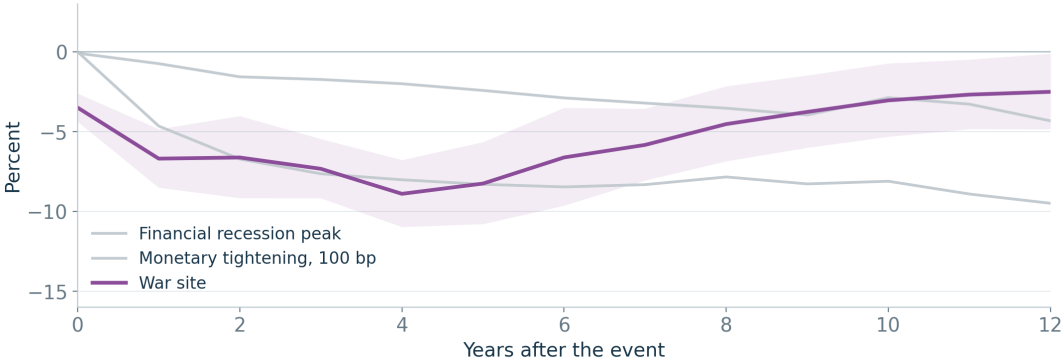
Monetary policy shocks have long run effects



A 100 bp tightening: GDP is 4.3% lower at year twelve.

Jordà–Singh–Taylor, REStat (2026): The long-run effects of monetary policy.

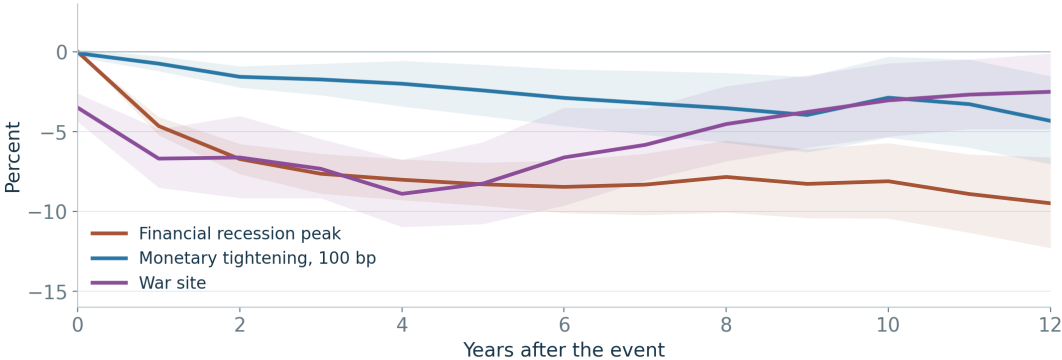
The Price of War



A deep initial loss, followed by substantial recovery.

Federle Meier Muller Mutschler & Schularick "The Price of War." War site with a 2% casualty/population ratio.

Disasters everywhere!



Growing evidence of persistent losses from various events

Jordà Schularick & Taylor *Disasters Everywhere...* (2024) IMF Economic Review

A simple log-utility welfare calculation

$$W = E\sum_{t=0}^{\infty}\beta^t \log C_t, \qquad \beta = 0.96, \quad p = 0.03.$$

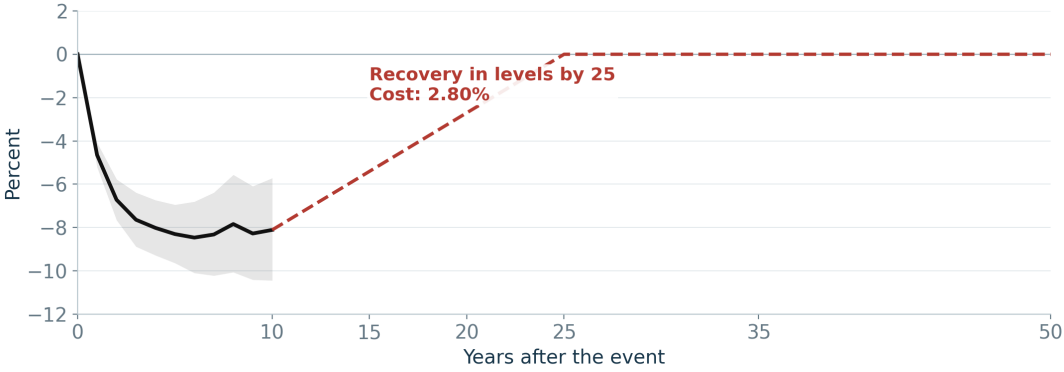
$$W^0 - W = \frac{p}{1-\beta}\sum_{h=0}^{\infty}\beta^h(-\theta_h) = -\frac{\log(1-\kappa)}{1-\beta}.$$

$$\boxed{\kappa = 1 - \exp\left[-p\sum_{h=0}^{\infty}\beta^h(-\theta_h)\right]}$$

100κ: **permanent percent of consumption we would give up to avoid crises.**

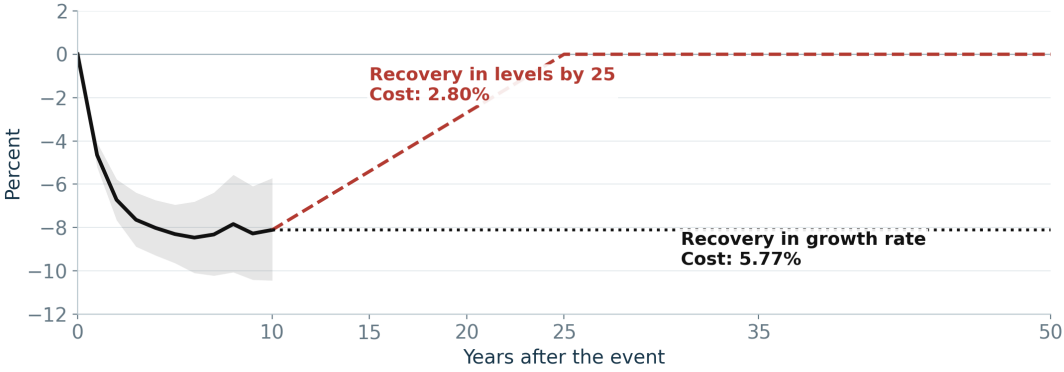
θ_h: estimated log response from the LP. GDP proxies consumption. Annual crisis probability p set at 3% throughout.

Why the recovery path matters



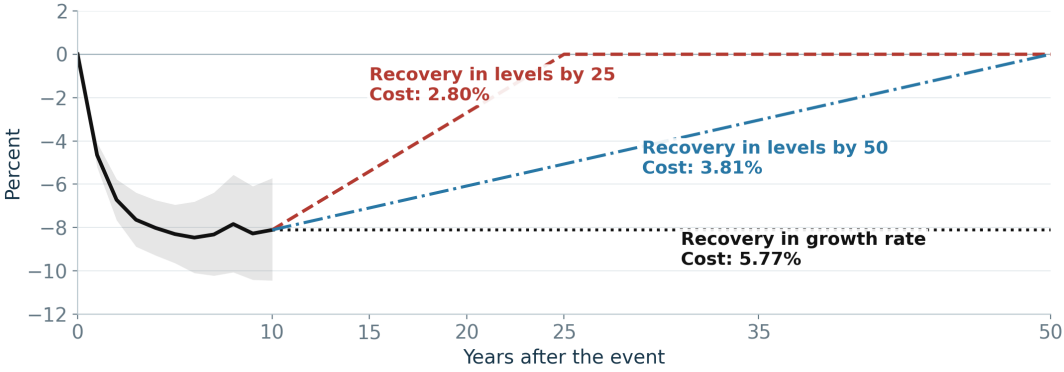
JST estimated through year 10. Welfare loss: 100κ; $p = 3\%$, $\beta = .96$.

Why the recovery path matters



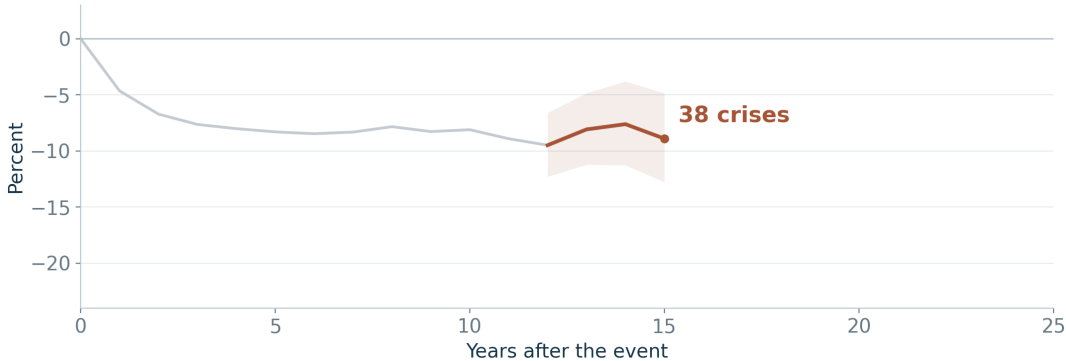
JST estimated through year 10. Welfare loss: 100κ; $p = 3\%$, $\beta = .96$.

Why the recovery path matters



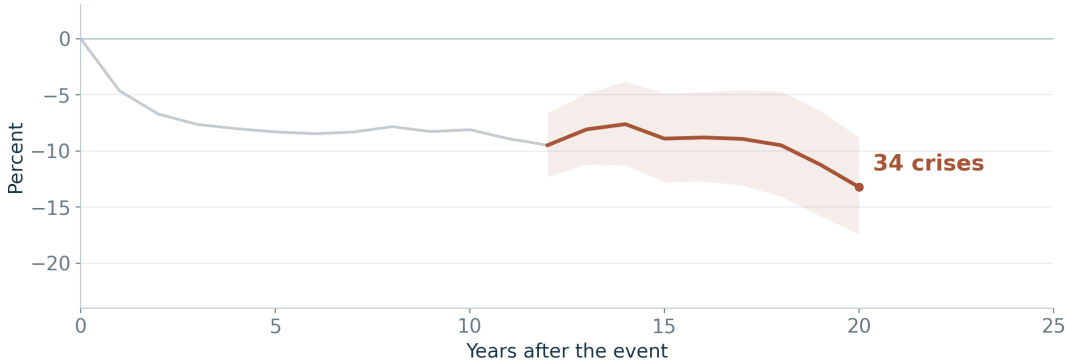
JST estimated through year 10. Welfare loss: 100κ; $p = 3\%$, $\beta = .96$.

The JST extended through year 15



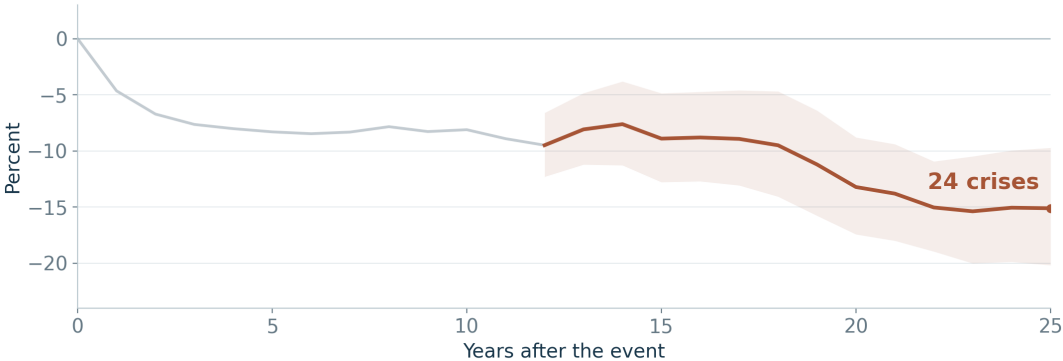
LP; four growth lags; country effects. Horizon-specific samples. 90% country-cluster intervals.

The JST extended through year 20



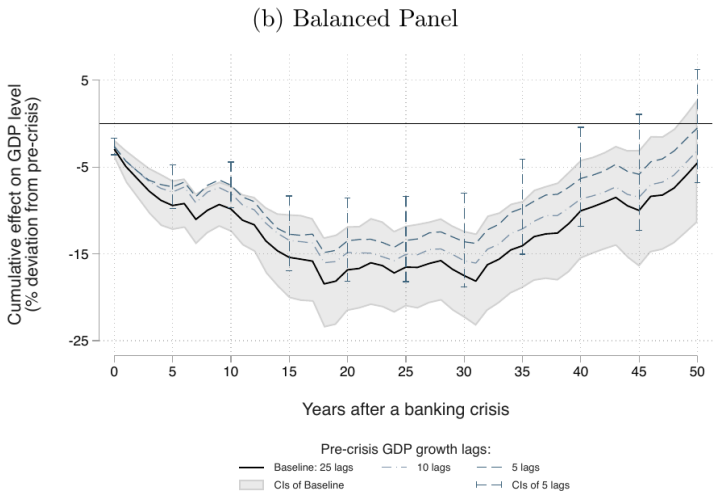
LP; four growth lags; country effects. Horizon-specific samples. 90% country-cluster intervals.

The JST extended through year 25



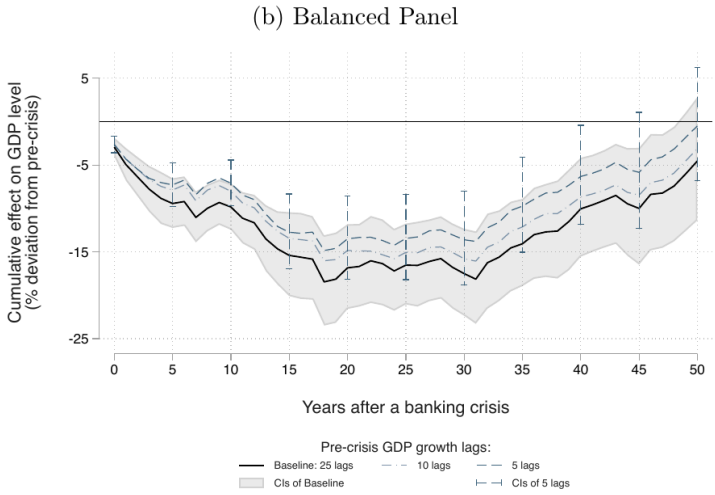
LP; four growth lags; country effects. Horizon-specific samples. 90% country-cluster intervals.

GMD has lot more events and histories



MXL panel
$>$ 130 crises

GMD has lot more events and histories

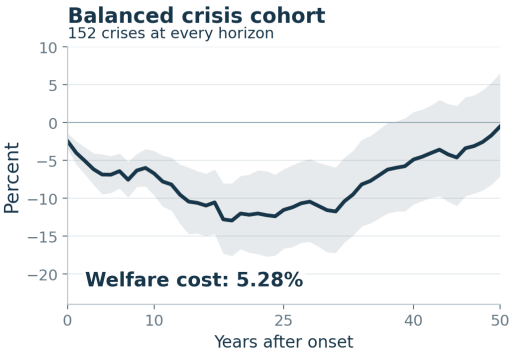


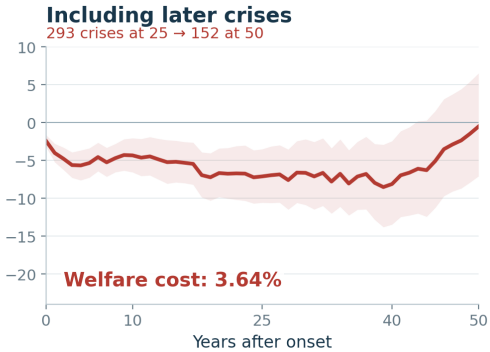
MXL, Figure 3(b).

MXL panel
$>$ 130 crises

Implied welfare cost:
about 5% of consumption.

Welfare costs depend on the sample, but remain large





MXL panel. Same controls at each horizon.

Does more data permit inference at longer horizons?

Longer histories let us trace recovery over longer horizons.

But are conventional confidence intervals reliable when h is large relative to T, especially with highly persistent output?

Example: the random walk

Suppose we observe the shock ε_t (& no intercept):

$$y_t = y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2).$$

Example: the random walk

Suppose we observe the shock ε_t (& no intercept):

$$y_t = y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2).$$

Local projection regression

$$y_{t+h} - y_{t-1} = \beta_h \varepsilon_t + u_{t,h}.$$

The true response is $\beta_h = 1$ at every horizon.

Why long horizons are noisy

In the random-walk example, the true response is $\beta_h = 1$: the original shock permanently raises output.

$$\widehat{\beta}_h - 1 = \frac{\sum_{t=1}^{T-h} \varepsilon_t \left(\sum_{j=1}^{h} \varepsilon_{t+j} \right)}{\sum_{t=1}^{T-h} \varepsilon_t^2} = O_p\left(\sqrt{\frac{h}{T-h}} \right).$$

Why long horizons are noisy

In the random-walk example, the true response is $\beta_h = 1$: the original shock permanently raises output.

$$\widehat{\beta}_h - 1 = \frac{\sum_{t=1}^{T-h} \varepsilon_t \left(\sum_{j=1}^{h} \varepsilon_{t+j} \right)}{\sum_{t=1}^{T-h} \varepsilon_t^2} = O_p\left(\sqrt{\frac{h}{T-h}} \right).$$

- **Numerator:** Did the original shocks happen to be followed by favorable or unfavorable subsequent shocks?
- **Denominator:** How much variation in the original shocks do we have to identify their effect?

Precision depends on the $h/(T-h)$. Increasing both proportionately can use up the gain.

Long-horizon confidence intervals can under-cover

Montiel Olea & Plagborg-Møller (2021), Table I:

Unit-root AR(1), $T = 240$. Actual coverage of nominal 90% intervals:

h	h/T	LP-LA bootstrap-t	LP-LA normal/EHW
12	0.05	84.3%	67.6%
36	0.15	74.1%	42.8%
60	0.25	64.2%	27.6%

Long-horizon confidence intervals can under-cover

Montiel Olea & Plagborg-Møller (2021), Table I:

Unit-root AR(1), $T = 240$. Actual coverage of nominal 90% intervals:

h	h/T	LP-LA bootstrap-t	LP-LA normal/EHW
12	0.05	84.3%	67.6%
36	0.15	74.1%	42.8%
60	0.25	64.2%	27.6%

Many countries help — but only if their crises are independent. Crises come in waves.

Long-horizon confidence intervals can under-cover

Montiel Olea & Plagborg-Møller (2021), Table I:

Unit-root AR(1), $T = 240$. Actual coverage of nominal 90% intervals:

h	h/T	LP-LA bootstrap-t	LP-LA normal/EHW
12	0.05	84.3%	67.6%
36	0.15	74.1%	42.8%
60	0.25	64.2%	27.6%

Many countries help — but only if their crises are independent. Crises come in waves.

Suggestions:

- Check long-horizon coverage using the panel's historical structure and international comovement.
- Show how confidence intervals change when allowing for international comovement, using Driscoll–Kraay standard errors.

Conclusion

I enjoyed reading this paper.

The data collection is a major contribution.

It lets us study crisis losses and recovery over much longer horizons.